

Mini Project Task 1

Real-Time ECG Acquisition, Plotting, R-Peak Detection, Heart Rate, and HRV in MATLAB

Student instruction sheet prepared for mini project implementation

Objective: Interface an ECG sensor with either Raspberry Pi or ESP32/Arduino, acquire ECG data in real time, send it to MATLAB, display a clean live ECG waveform, plot one clean ECG cycle, detect R-peaks, calculate heart rate (BPM) every 10 seconds, and calculate HRV (ms) after one minute.

1. Hardware options

Option	Available hardware	Important note
A	Raspberry Pi available in the DSP Lab	Raspberry Pi has no analog input pins, so an ADC is required before reading ECG sensor output.
B	ESP32 or Arduino available in the Embedded Systems Lab or other SEECS labs	These platforms can read analog ECG sensor output directly without an extra ADC in many common setups.

The ECG sensor may either be provided by the teacher or purchased by the students. The teacher may contribute some amount toward the purchase (only for ECG sensor); this will be confirmed separately by the teacher. You can purchase ecg sensor from any online shop or from college road Rawalpindi. One link is given below for your information.

Electrobes ECG sensor: <https://electrobes.com/product/ecg-module-ad8232-heart-ecg-monitoring-sensor-module-kit-for-arduino/>



2. Core task requirements

- Interface the ECG sensor with any one hardware platform: Raspberry Pi with ADC, or ESP32, or Arduino.
- Acquire ECG data in real time and send it to MATLAB using a suitable communication method.
- Plot the ECG waveform in real time. The waveform must look like an actual ECG signal, not a noisy or meaningless garbage waveform.
- On the same figure or dashboard, also show one clean ECG cycle.
- Detect the R-peaks during real-time operation.
- Calculate heart rate in beats per minute (BPM) after every 10 seconds of incoming data.
- Calculate HRV in milliseconds after one minute of data.
- The main processing, plotting, peak detection, heart rate, and HRV calculations will be performed in MATLAB.

3. MATLAB connectivity guidance

Students can search and select any suitable method for connecting Raspberry Pi, ESP32, or Arduino to MATLAB. Possible approaches include:

- Serial communication through a COM port.
- Wi-Fi-based communication.
- TCP/IP communication between the acquisition device and MATLAB.
- Using MATLAB support packages or hardware toolboxes where already available.

A practical workflow is to acquire ECG data on Raspberry Pi, ESP32, or Arduino, then stream that data to MATLAB. Port definition and connection setup are usually simple and may require only a minimal number of lines of code on each side, depending on the chosen method.

4. Expected display layout

The figure below shows the expected MATLAB display concept: a real-time ECG panel, a one-cycle ECG panel, and a summary box for BPM and HRV updates.

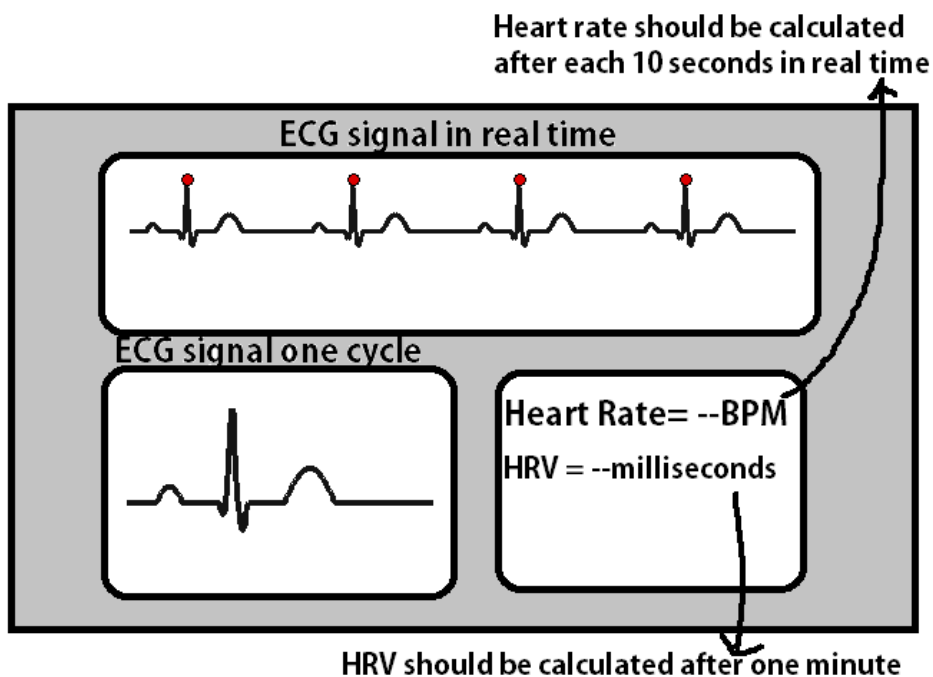


Figure 1. Suggested Task 1 dashboard layout with ECG waveform placeholders filled.

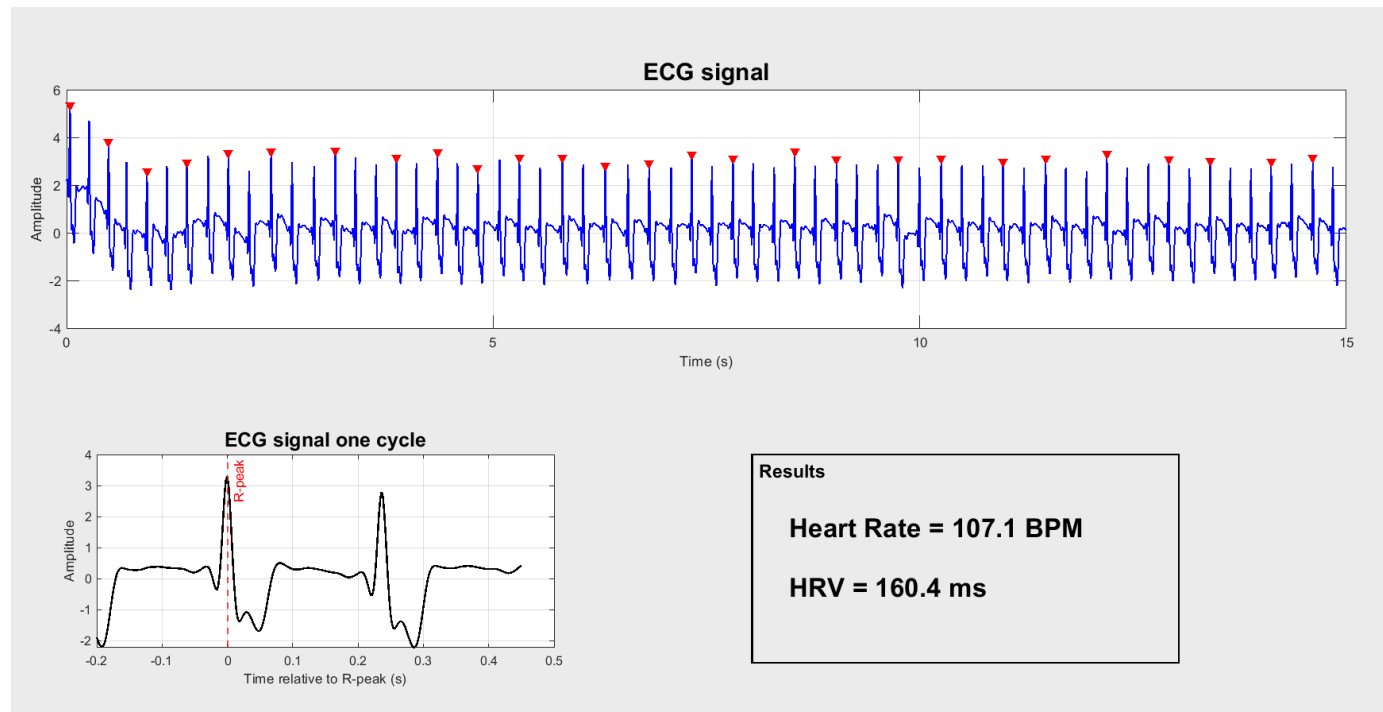


Figure 2. Suggested Task 1 dashboard layout with ECG waveform placeholder using example.

5. Deliverables from each student group

- Working hardware setup with sensor interfaced to one selected platform.
- MATLAB code for acquisition, plotting, R-peak detection, BPM calculation, and HRV calculation.
- Live demonstration of real-time ECG plotting.
- A brief explanation of the selected communication method between hardware and MATLAB.
- A clean plot or screenshot clearly showing ECG waveform quality, one cycle, BPM, and HRV.
- Additional Submission file will be created on LMS.

6. Notes from the instructor

- Signal quality matters. Students must ensure the plotted waveform represents a true ECG pattern.
- Students may issue the required hardware from the relevant lab for mini project purposes, subject to lab rules.
- Final confirmation regarding ECG sensor provision or teacher support amount (only for sensor) will be communicated by the teacher.
- Students are encouraged to search for the most suitable MATLAB-device connectivity approach instead of limiting themselves to a single method.